IMPLEMENTATION OF FASTSPEECH 2-BASED TEXT-TO-SPEECH SYNTHESIS

TECHNICAL REPORT

1. INTRODUCTION

This project implements two independent FastSpeech 2 text-to-speech systems for English and Telugu. The systems use the Svarah English dataset and IndicTTS Telugu dataset respectively.

FastSpeech 2 is a non-autoregressive neural TTS architecture that uses duration, pitch and energy information to improve speech generation and prosody control.

2. OBJECTIVES

- Implement FastSpeech 2 for English and Telugu.
- Prepare language-specific training and validation datasets.
- Extract mel-spectrogram, pitch, energy and duration features.
- Train separate models for both languages.
- Generate speech from the prescribed test sentences.
- Visualize variance adaptor predictions.

3. DATASET PREPARATION

ENGLISH:

Dataset: AI4Bharat Svarah.

Total available recordings: 6,656.

Training duration: approximately 5 hours.

Validation duration: approximately 1.006 hours.

Training recordings: 3,449.

Validation recordings: 695.

Sampling rate after preprocessing: 22,050 Hz.

TELUGU:

Dataset: AI4Bharat IndicTTS Telugu.

Training duration: approximately 5.0013 hours.

Validation duration: approximately 1.0003 hours.

Training recordings: 2,036.

Validation recordings: 376.

Sampling rate: 22,050 Hz.

Training and validation sets were separated without overlap.

Preprocessing included audio resampling, text processing,

forced alignment and extraction of acoustic features.

4. MODEL ARCHITECTURE

The implementation is based on the FastSpeech 2 repository:

https://github.com/ming024/FastSpeech2

The architecture contains:

- Text encoder
- Variance adaptor
- Duration predictor
- Pitch predictor
- Energy predictor
- Mel-spectrogram decoder
- PostNet
- HiFi-GAN vocoder

The variance adaptor uses duration, pitch and energy predictions

to condition mel-spectrogram generation.

5. LANGUAGE-SPECIFIC MODIFICATIONS

ENGLISH:

The English system uses English text cleaners and phoneme

conversion. The Svarah dataset was processed with forced alignment

and phoneme-level acoustic features.

TELUGU:

The Telugu implementation includes:

- Telugu Unicode NFC normalization.
- Telugu-specific text cleaning.
- A Telugu symbol inventory.
- Grapheme-level text processing.
- Special handling of the spn token.
- Telugu-specific inference preprocessing.

Separate vocabulary inventories are preserved for English and Telugu because their trained checkpoints use different input vocabularies.

6. TRAINING CONFIGURATION

Both models were trained for 10,000 steps.

ENGLISH FINAL TRAINING LOSSES:

Total loss: 3.2916

Mel loss: 0.8888

Mel PostNet loss: 0.8897

Pitch loss: 0.7400

Energy loss: 0.3361

Duration loss: 0.4368

ENGLISH FINAL VALIDATION LOSSES:

Total loss: 4.4079

Mel loss: 1.0367

Mel PostNet loss: 1.0367

Pitch loss: 1.1540

Energy loss: 0.9282

Duration loss: 0.2524

TELUGU FINAL TRAINING LOSSES:

Total loss: 1.6960

Mel loss: 0.6714

Mel PostNet loss: 0.6712

Pitch loss: 0.1253

Energy loss: 0.1509

Duration loss: 0.0772

TELUGU FINAL VALIDATION LOSSES:

Total loss: 4.3766

Mel loss: 1.0917

Mel PostNet loss: 1.0920

Pitch loss: 1.2007

Energy loss: 0.8314

Duration loss: 0.1609

The English training log history was unavailable after a Kaggle session reset. Its recorded final loss values are retained, but the original English loss curves could not be reconstructed. The Telugu training and validation loss curves are included in the results directory.

7. INFERENCE RESULTS

Five prescribed test samples were synthesized for each language.

English:

E1, E2(a), E2(b), E3 and E4.

Telugu:

T1, T2(a), T2(b), T3 and T4.

The generated audio files are included in:

Results/English/test_generated_audios/

Results/Telugu/test_generated_audios/

The original supplied test sentences were retained during testing.

8. VARIANCE ADAPTOR ANALYSIS

For each test sentence, duration, pitch and energy prediction plots were generated.

Each language contains:

- Five duration plots.
- Five pitch plots.
- Five energy plots.

These plots help inspect predicted speaking duration, pitch variation and energy variation across the input sequence.

Pitch and energy plots represent model prediction values and should not automatically be interpreted as physical Hz or amplitude measurements.

9. COMPARISON AND OBSERVATIONS

Both language-specific models completed 10,000 training steps and generated five test audio samples.

The English final training total loss was 3.2916, while its validation total loss was 4.4079.

The Telugu final training total loss was 1.6960, while its validation total loss was 4.3766.

The difference between training and validation losses suggests that generalization remains an area for improvement, particularly for Telugu.

The generated audio and variance adaptor plots provide qualitative material for evaluating pronunciation, prosody and prediction behaviour. Listening-based evaluation is necessary before making strong claims about speech naturalness or intelligibility.

10. LIMITATIONS

- The datasets contain limited training durations compared with large-scale production TTS systems.
- The Telugu training and validation loss gap indicates possible overfitting.
- English loss history was not available after the session reset.
- No formal MOS or subjective listening evaluation was conducted.
- Pronunciation quality may vary for unusual words, numbers, punctuation and malformed input spacing.
- Reproducibility requires the corresponding datasets and external alignment resources.

11. CONCLUSION

Two language-specific FastSpeech 2 systems were implemented for English and Telugu. Dataset preparation, feature extraction, model training, inference and variance adaptor visualization were completed.

The project produced ten synthesized test audio samples and thirty variance adaptor plots across both languages.

The final checkpoints, configurations, preprocessing features, source code and generated results are provided with the project.

12. REFERENCES

1. Ren et al., FastSpeech 2: Fast and High-Quality End-to-End

Text to Speech, ICLR 2021.

2. FastSpeech 2 implementation:

https://github.com/ming024/FastSpeech2

3. Svarah dataset:

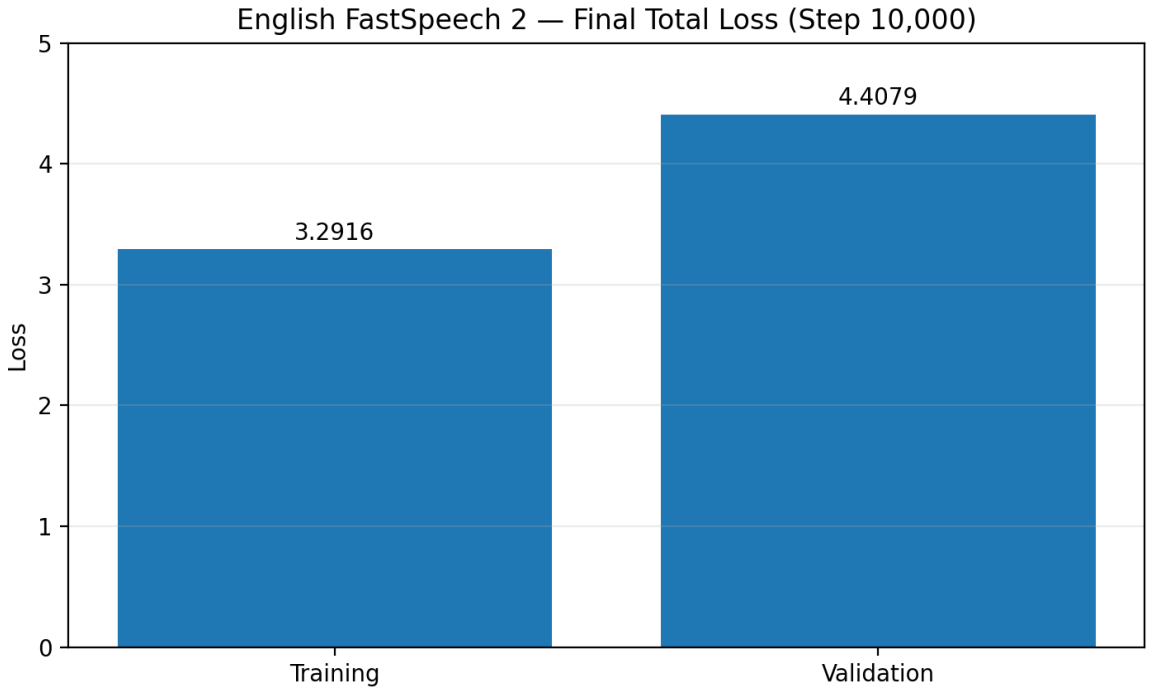
https://huggingface.co/datasets/ai4bharat/Svarah

4. AI4Bharat IndicTTS:

https://github.com/AI4Bharat/Indic-TTS

Appendix: English FastSpeech 2 Final Loss Summary

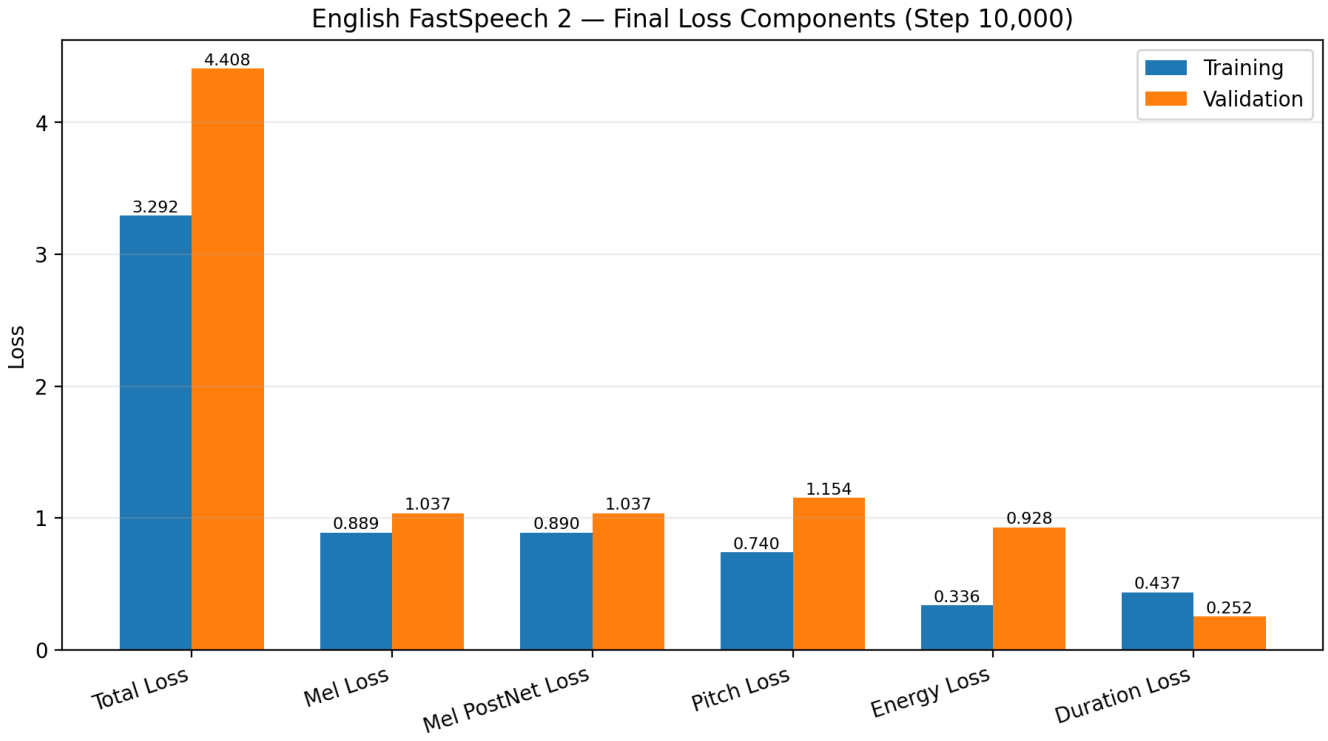
These visualizations summarize the recorded training and validation losses at step 10,000. They are final-step comparisons, **not training-progress curves**.



Final loss components

Metric	Training	Validation
Total Loss	3.2916	4.4079
Mel Loss	0.8888	1.0367
Mel PostNet Loss	0.8897	1.0367
Pitch Loss	0.7400	1.1540
Energy Loss	0.3361	0.9282
Duration Loss	0.4368	0.2524

Comparison of Individual Loss Components



Data provenance: Values are taken from the supplied English_loss_summary.txt, which records the final output at training step 10,000. The original English training and validation log histories were unavailable after the Kaggle session reset. No intermediate 10,000-step values have been estimated or fabricated.

Interpretation: The final validation total loss (4.4079) is higher than the final training total loss (3.2916). This endpoint comparison does not show how either loss changed during training and should not be interpreted as a loss curve.